TOKTAM SAMIEI

Machine Learning • NeuroAI • Computational Neuroscience • Algorithm Development • Signal Processing • Data Science

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Professional Profile

Machine learning researcher, computational neuroscientist, and Ph.D. candidate specializing in neural time-series analysis, neural decoding, and spatiotemporal modeling. Develops algorithms and interpretable models to investigate how neural populations represent, integrate, and transmit information. Experience includes Transformer-based decoding and neural activity prediction, biologically inspired spiking neural networks, and multimodal learning, supported by GPU computing, ablation studies, and statistical validation.

Technical Skills

Programming and libraries: Python, MATLAB, C++; NumPy, SciPy, pandas, scikit-learn, Matplotlib.

Deep learning: PyTorch, snnTorch, TensorFlow/Keras; Transformers, spike-event tokenization, spatiotemporal embeddings, CNNs, RNNs, spiking neural networks (LIF/RLIF), deep canonical correlation analysis.

Machine learning and predictive modeling: Regularized regression, autoregressive models, Gaussian-kernel regression, nonlinear model fusion, random forests, PCA, hyperdimensional computing, multimodal learning.

Neural data and signal processing: Neuropixels spike trains, EEG/iEEG, LFP, calcium imaging, eye tracking; neural and behavioral decoding, multivariate time-series analysis, adaptive temporal segmentation, spectral analysis, filtering, phase locking, signal and noise correlations; AllenSDK, EEGLAB.

Model interpretation and validation: Attention analysis, ablation studies, synthetic neural data generation, ground-truth connectivity benchmarks, randomized controls, hyperparameter sensitivity analysis, statistical inference, information theory.

Scientific computing: GPU computing with PyTorch and CuPy; SLURM, joblib, multiprocessing; parallel parameter sweeps, reproducible training and analysis pipelines.

Research and Technical Experience

Graduate Research Assistant | University of California, Riverside | Sep 2022–Present

- **Transformer-based neural causal discovery:** Developing a spatiotemporal attention-based Transformer for self-prediction of neural spiking activity to identify directed neuronal interactions and validate attention-guided causal connectivity through ablation studies against ground-truth synthetic networks.
- **Multi-region neural decoding:** Developed region-specific Transformer models to investigate synergistic versus additive contributions to behavioral decoding from simultaneous multi-region neural recordings, using self-attention to model within-region dynamics and cross-attention to assess gains beyond linear fusion of independent nonlinear representations.
- **Adaptive temporal modeling:** Designed Integration–Split, a greedy temporal-segmentation algorithm guided by regularized logistic regression, to identify dynamic optimal timescales (the speed of information processing in the brain) in spiking neural activity.
- **Transformer decoding and interpretability:** Implemented a PyTorch model for visual stimulus decoding from sparse spike-event sequences, with neuron-identity and continuous-time embeddings. Built stratified training, validation, and test workflows and validated learned attention against optimal timescale estimates, providing evidence that learned attention weights can reflect biological dynamics. Implemented attention-head ablation to quantify the functional relevance of each head. Evaluated shuffled-label, shuffled-spike, and untrained controls to distinguish learned temporal structure from baseline effects.
- **Predictive modeling:** Combined autoregressive and Gaussian-kernel predictors using linear and gated nonlinear fusion. Compared held-out variance explained to characterize when nonlinear interactions between neural history and stimulus input improve prediction.
- **Neural representation learning:** Developed NeuroPixelHD, a hyperdimensional decoder for visual stimuli. Benchmarked against random forests, kNN, and neural networks to evaluate spatial and temporal aggregation strategies.
- **Population signal analysis:** Developed sweeping-integration and statistical analysis pipelines using Allen Institute and IBL recordings, leading to the identification of bracket coding, a hybrid coding mechanism in the brain that integrates rate and temporal coding. Assessed temporal boundaries, cross-region synchrony, and LFP phase locking against randomized controls. Implemented post hoc analyses to validate the coding mechanism across the sensory hierarchy.

- **Spiking neural network modeling:** Developed and trained LIF and recurrent LIF networks in PyTorch/snnTorch to investigate the generation and decoding of bracket-coded spike trains, comparing rate-based and spike-timing training objectives. Used ridge regression, input-removal controls, and hyperparameter sensitivity analyses to assess how recurrent dynamics and auxiliary signals support temporal encoding and class discrimination.
- **Scientific software:** Built CPU/GPU analysis workflows and SLURM batch jobs for neural preprocessing, decoding, and parameter sweeps; implemented reusable analysis modules and training-derived centering for evaluation data.

Research Assistant | University of Tehran | Sep 2021–Apr 2022

- Led a multimodal emotion-recognition project combining EEG and eye-movement data with deep canonical correlation analysis to learn shared representations across modalities.

Research Intern | University of Tehran | Jun–Sep 2020

- Investigated ultrasound-induced refractive-index modulation for guiding light through scattering tissue in research on non-invasive deep-brain stimulation.

Education

Ph.D. in Mechanical Engineering, University of California, Riverside | Expected June 2027

Advisor: Prof. Erfan Nozari | Computational neuroscience, neural decoding, and machine learning

M.S. in Mechanical Engineering, University of California, Riverside | Jun 2024

Advisor: Prof. Erfan Nozari

B.S. in Engineering Science, University of Tehran | Apr 2021

Electrical Engineering Area of Study | Ranked 2nd

Publications and Manuscripts

Peer-Reviewed Publications

Ahmed, H. F., **Samiei, T.**, & Nozari, E. (2026). On the Optimal Temporal Resolution for Information Representation in Neural Activity: A Theoretical Analysis. *Frontiers in Computational Neuroscience*, 20, 1885975.

doi:10.3389/fncom.2026.1885975

Samiei, T., Zou, Z., Imani, M., & Nozari, E. (2024). Optimal Decoding of Neural Dynamics Occurs at Mesoscale Spatial and Temporal Resolutions. *Frontiers in Cellular Neuroscience*, 18, 1287123.

doi:10.3389/fncel.2024.1287123 | Research code: NeuroPixelHD

Preprint

Samiei, T., Ahmed, H. F., Zagha, E., & Nozari, E. (2026). Bracket Coding: An Emergent Balance Between Temporal Integration and Segregation in Early Visual Population Activity. *bioRxiv*.

Preprint | Research code: Bracket Coding

Samiei, T., Lokray, S., Ahmed, H. F., & Nozari, E. (2026). Dynamic Optimal Temporal Resolution: Reflecting the Evolving Population-Level Characteristics.

Preprint | Research code: Dynamic Optimal Temporal Resolution

Presentations,Training and Academic Service

Conferences: Society for Neuroscience (SfN) Annual Meeting: posters presented in 2023, 2024 and 2025; poster accepted for 2026. Selected to participate in the Lake Conference on Neural Coding and Dynamics.
Neuromatch Academy, Computational Neuroscience Summer School | 2022

Intensive training in neural modeling and data analysis, including a collaborative research project.

Selected coursework: Artifitial Inteligence, Machine Learning; Deep Learning; Reinforcement Learning, COmputer Vision, Digital Signal Processing; Linear Algebra; Dynamical Systems; Optimal Control and Estimation; Cellular Neuroscience.

Teaching: Linear Control Systems (four offerings), Dynamical Systems, Electromagnetics, and Physics II (two offerings).

Ad hoc reviewer: *Nature Communications*, *Communications Physics*, *Chaos*, and *iScience* (2026).